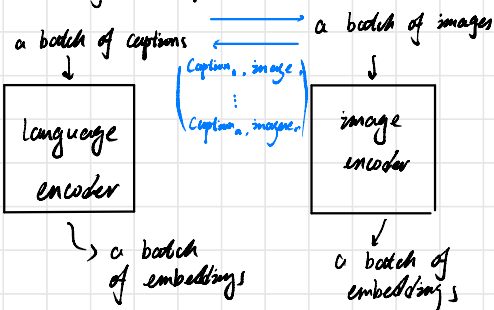
The core idea of contrastive learning is simple and intuitive:

Similar things should be grouped together, while dissimilar things should be pushed apart.

Example: Structure for CLIP



$$S = \begin{pmatrix} s_{1,1} & \cdots & s_{1,n} \\ \vdots & \ddots & \vdots \\ s_{n,1} & \cdots & s_{n,n} \end{pmatrix}$$

Loss function (Symmetric Cross-entropy Loss)

Image to text loss:

$$\mathcal{L}_i^{(I \to T)} = -\log \frac{\exp(s_{i,i}/\tau)}{\sum_{j=1}^{n} \exp(s_{i,j}/\tau)}$$

→ this is a learnable parameter. a good learned temperature can reflect the true probability → lead to good expected calibration error

→ this part can be interpreted as probability (the probability of image i to find the correct text)

Similarly, we can have

Text to image loss:

$$\mathcal{L}_i^{(T \to I)} = -\log \frac{\exp(s_{i,i}/\tau)}{\sum_{j=1}^{n} \exp(s_{j,i}/\tau)}$$

Final symmetric loss:

$$\sum_{i=1}^{n} \frac{\left(\mathcal{L}_i^{(I \to T)} + \mathcal{L}_i^{(T \to I)}\right)}{2n}$$

Other Contrastive loss methods:

pairwise Contrastive loss:

This method looks at one pair of samples at a time. Each pair is labeled as either positive (image i with caption i) or negative. (image i with caption j)

$$\text{loss} = Y_i D^2 + (1 - Y_i) \cdot \max(0, m - D)^2$$

- Y_i: if positive $Y_i = 1$, otherwise $Y_i = 0$
- $Y_i D^2$: if the pair is positive, minimize D
- m: margin
- $(1 - Y_i) \cdot \max(0, m - D)$: if the pair is negative, push the distance to more than m, but if the distance is already larger than m, leave it as it is.

The idea of pairwise contrastive loss is simple, if the pair matches, push the distance to 0. if however the pair does not match, push the distance large.

However, once the distance is larger than 0, the loss = 0. Why is this?

1. If there is no such margin, the model could push one mismatch pair to infinite, destroying the loss function.

2. This forces the model to update its weights from positive pairs and "hard negative" pairs. Since "easy negative" pairs easily have distance > m, they do not provide any gradients.

triplet loss:

look at three data points at a time

1. Anchor (a): the baseline image
2. Positive (p): paired caption
3. Negative (n): a unpaired caption

$$\text{loss}: \max(0, d(a,p) - d(a,n) + m)$$

- m: again we would have an margin here for similar reasons as in pairwise contrastive loss
- $d(a,p) - d(a,n)$: the distance of a and p should be smaller than "a to" n by at least m. (it only cares about relative distances)

Compare three loss functions:

